Daily Number Theory Drill #3

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Cyclic Patterns · Telescoping Exponents · Last Digit Sequences.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition (≤15 s each)

Remainders, parity, basic prime factors, and pure recognition.

1. Find the remainder when $10^{10} + 10^5 + 1$ is divided by 9.

2. What is the unit digit of 3^{2026}?

3. Find the largest prime factor of 1001.

4. Find the remainder when $7 \times 8 \times 9$ is divided by 5.

5. If n is an integer such that $2n \equiv 3 \pmod 5$, what is the smallest positive value of n?

6. Evaluate $\gcd(144, 84)$.

7. Find the number of positive divisors of 36.

8. Find the remainder when 2^{10} is divided by 11.

9. Express 120 as a product of prime factors.

10. Find the smallest positive integer k such that $12k$ is a perfect square.

Section B Manipulation Drills (≤50 s each)

Divisibility rules, modulo properties, algebraic factoring over integers.

1. How many of the numbers $3, 4, 5, 6, 7$ are factors of the sum $3^{2026} + 3^{2025} + 3^{2024}$? *(after SMC 2023 Q10)*

2. Find the highest power of 2 that divides 100!.

3. If a and b are non-zero digits, find the maximum possible integer value of $\dfrac{10a + b}{a + b}$.

4. Find the number of integer pairs (x, y) such that $xy = 24$.

5. What is the smallest positive integer greater than 2 that leaves a remainder of 2 when divided by $3, 4$, and 5?

6. Find the remainder when $11^{11} + 12^{12}$ is divided by 10.

7. Find the value of p if p and $p^2 + 2$ are both prime numbers.

8. Given that n is an integer, for how many values of n is $\dfrac{2n + 15}{n + 2}$ an integer?

9. Find the sum of all positive integers n for which $n^2 - 1$ is a prime number.

10. Determine the two last digits of $5^{2026} + 6^{2026}$.

Section C Substitution & Structure

(≤90 s each)

Link Number Theory to Algebra and Combinatorics. Look for patterns.

1. Find all pairs of positive integers (x, y) such that $\dfrac{1}{x} + \dfrac{1}{y} = \dfrac{1}{3}$.

2. Solve the equation $x^2 - y^2 = 2025$ for positive integers (x, y) such that y is as small as possible.

3. Find all pairs of integers (x, y) that satisfy $xy + x + y = 14$.

4. Find all integers x such that $x^4 + x^2 + 1$ is a prime number.

5. How many ways can we choose the digits a and b such that the 5-digit number $3a4b2$ is divisible by 36?

6. Find the number of 3-element subsets of $\{1, 2, \ldots, 10\}$ such that the sum of the elements is divisible by 3.

7. Find the sum of all integer values of x for which $\dfrac{x^2 + 7x + 2}{x + 1}$ is an integer.

8. Find the last two digits of 2019^{2025}.

Section D Challenge

(TMUA / SMC / BMO1 difficulty)

Insight over computation. Each question has a short, elegant solution — find it.

1. Find the number of pairs of positive integers (a, b) with $a \leq b$ such that $\text{lcm}(a, b) = 2^3 \cdot 3^2 \cdot 5$.

2. Find all pairs of prime numbers (p, q) such that $p^2 - 2q^2 = 1$. *(after BMO1 2010 Q1)*

3. Find the number of integer solutions to $x^3 + 2y^3 = 4z^3$. *(after BMO1 2014 Q1)*

4. Find the last three digits of 7^{9999}. *(after MAT 2014 Q1F)*

5. Find the last three digits of 2025^{2026}. *(after SMC 2023 Q25)*

"Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised."

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths